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Marks: 1

Let X and Y have joint probability density function(pdf) $f(x,y) = ce^{-5y}$, $0 < x < y < \infty$ and zero otherwise. Find the mean of the conditional distribution of $Y|X=3.85$. _____

Answer:

2 

Marks: 1

Let X_1 and X_2 be independent random variables. X_1 follows a gamma distribution with parameters $\alpha = 2$ and $\beta = 1/10$, whereas X_2 follows an exponential distribution with mean $1/6$. Find $E[X_1|X_1 + X_2 = 4]$. _____

Answer:

3

Marks: 1

On an auto collision coverage, there are 3 classes of policyholders, A, B and C. 40% of drivers are in class A, 27% of the drivers are in class B and 33% in class C. The distribution of losses for the drivers are:

Class	Losses		
	1000	5000	10000
	Probability		
A	0.47	0.26	0.27
B	0.48	0.32	0.2
C	0.47	0.33	0.2

A claim is submitted by a randomly selected driver. Calculate the standard deviation of the size of the claim. _____

Answer:

4

Marks: 1

Given a value of $(\Theta = \theta)$, the random variable X follows a Gamma distribution with probability density function

$$f(x) = \theta^2 x e^{-\theta x}.$$

Θ has a uniform distribution on the interval $(3, 8)$. Determine $S_X(0.31)$ for the unconditional distribution. _____

Answer:

5

Marks: 1

Annual claim counts per risk are binomial with parameter $m = 3$ and Q . Q varies by risk uniformly on $(0.28, 0.78)$. For a risk selected at random, determine the probability of at most one claims. _____

Answer:

6

Marks: 1

You are given:

- Conditional on q , the random variables $X_1, X_2, \dots, X_m$, are independent and follow a Bernoulli distribution with parameter q .

- $S_k = X_1 + X_2 + \dots + X_k$.
- The distribution of q is Beta($a=2$, $b = 2$, $\theta = 12$).

Determine the variance of the marginal distribution of S_{113} . _____

Answer:

7 

Marks: 2

The size of loss , X has mean 4λ and variance $7\lambda^2$. Λ has the following density function:

$$f(\lambda) = 5(1250)^5/(\lambda+1250)^6.$$

Calculate the variance of the loss. _____

Answer:

8 

Marks: 2

The annual number of accidents for an individual driver has a Poisson distribution with mean λ . The mean, Λ , of a heterogeneous population of drivers have a gamma distribution with mean 0.46 and variance 0.1058. Calculate the probability that a driver selected at random from the population will have 2 or more accidents in one year. _____

Answer:

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